Lecture 8 Compendium: Fundamentals of Spatial Information Science

Module 3: Technologies of GIS in Crop Protection

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Compendium for Advanced Study

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1 Introduction: From Mapping to Modeling

At the undergraduate level, Geographic Information Systems (GIS) are often presented as tools for visualizationmaking pretty maps. At the PhD level, we redefine GIS as **Spatial Information Science**. It is the rigorous mathematical and computational framework used to model the physical world.

In the context of **Agroecology** and **Crop Protection**, GIS is the engine that connects the "What" (a pest outbreak) with the "Where" (spatial location) and the "Why" (environmental covariates).

1.1 Tobler's First Law of Geography

The foundational axiom of all spatial analysis is Tobler's First Law (TFL):

> *"Everything is related to everything else, but near things are more related than distant things."*

Mathematically, this refers to **Spatial Autocorrelation**. If a Maize plant at location (x, y) has Fall Armyworm, the plant at $(x + \delta, y + \delta)$ has a high probability of also being infected. This statistical dependence violates the assumption of independence found in standard statistics (ANOVA/Regression), necessitating specific **Spatial Statistics** (e.g., Moran's I, Kriging).

2 Spatial Data Models: The Ontology of Place

To represent the infinite complexity of the real world in a finite digital system, we rely on two primary abstractions or "Data Models."

2.1 1. The Vector Model (Discrete Objects)

The vector model views the world as a collection of discrete objects with defined boundaries. It relies on **Coordinate Geometry**.

- **Point (0-D):** Defined by a single coordinate pair (x, y).
- **Line (1-D):** Defined by a sequence of points (vertices).
- **Polygon (2-D):** Defined by a closed sequence of lines where the first and last coordinates are identical: $(x_1, y_1) = (x_n, y_n)$.

2.1.1 Topology

In a PhD context, we do not just store "spaghetti lines." We enforce **Topology**the mathematical study of geometric properties that are preserved under deformation.

- **Adjacency:** Do two farm plots share a boundary?
- **Containment:** Is the pest trap *inside* the farm?
- **Connectivity:** Are the irrigation canals connected to the river?

2.2 2. The Raster Model (Continuous Fields)

The raster model views the world as a continuous surface. Space is tessellated into a regular grid of cells (pixels).

- **Structure:** A matrix $M_{i,j}$ where each cell holds a value (Digital Number, DN).
- **Use Case:** Continuous phenomena where "boundaries" are arbitrary, such as Elevation (DEM), Soil pH, Temperature, or Satellite Imagery (NDVI).

The Conversion Dilemma

Vector to Raster (Rasterization): Loss of precision. A fine boundary line becomes a jagged edge of pixels (aliasing). **Raster to Vector (Vectorization):** Creation of artifacts. Converting a gradient of soil moisture into discrete "High/Low" polygons loses the nuance of the transition zone. *PhD Rule:* Always perform analysis in the native format of the data source whenever possible.

3 Geodesy: The Mathematics of Earth

A common source of error in Tanzanian agricultural research is the misuse of Coordinate Reference Systems (CRS).

3.1 The Ellipsoid and Geoid

1. **The Geoid:** The equipotential surface of the Earth's gravity field (Mean Sea Level). It is "lumpy" because gravity varies.
2. **The Ellipsoid:** A smooth mathematical approximation (oblate spheroid) used for calculations. Defined by semi-major axis a and semi-minor axis b.

$$f = \frac{a - b}{a} \quad \text{(Flattening)}$$

3.2 The Datum Conflict: Arc 1960 vs. WGS 84

A **Datum** pins the mathematical ellipsoid to the physical earth.

- **WGS 84:** Global datum. Center of mass is Earth's center. Used by GPS.
- **Arc 1960:** Local datum for East Africa. "Best fit" for Tanzania/Kenya. Used in older government survey maps.

The Datum Shift: If you collect GPS points (WGS 84) and overlay them on a Ministry of Lands cadastral map (Arc 1960) without transformation, you will see a shift of ≈ 200m.

3.2.1 Coordinate Transformation (Python)

Using the 'pyproj' library to mathematically transform coordinates between datums.

```
from pyproj import Transformer

# Define the CRS (Coordinate Reference Systems)
# EPSG:4326  -> WGS 84 (Global GPS)
# EPSG:4210  -> Arc 1960 (Tanzania Survey)

```

```
wgs84_crs = "EPSG:4326"
arc1960_crs = "EPSG:4210"

# Create a transformer object
transformer = Transformer.from_crs(wgs84_crs, arc1960_crs)

# A sample GPS point in Morogoro (Lat, Lon)
lat_wgs = -6.832
lon_wgs = 37.664

# Transform
lat_arc, lon_arc = transformer.transform(lat_wgs, lon_wgs)

print(f"WGS 84:   {lat_wgs}, {lon_wgs}")
print(f"Arc 1960: {lat_arc:.5f}, {lon_arc:.5f}")
print("Note the shift in raw decimal degrees!")
```

Listing 1: Datum Transformation using Python

4 Map Projections: Flattening the Sphere

To measure area (hectares) or distance (meters), we must project the 3D earth onto a 2D plane.

4.1 Universal Transverse Mercator (UTM)

The standard for agricultural mapping. It divides the Earth into 60 zones, each 6° wide.

- **Properties:** Conformal (preserves shape/angles locally).
- **Unit:** Meters.

Tanzania's Zones:

- **Zone 35S:** Western (Kigoma).
- **Zone 36S:** Central (Dodoma, Mbeya).
- **Zone 37S:** Eastern (Morogoro, Dar es Salaam).

Scale Factor (k): In UTM, the scale is exact ($k = 1$) only on two secant lines. At the central meridian, $k = 0.9996$. This means a distance of 1000m on the ground measures 999.6m on the map. For high-precision cadastral surveys, this distortion must be corrected.

5 Spatial Databases: PostGIS

For PhD research involving large datasets (e.g., millions of yield points), desktop GIS (QGIS/ArcGIS) is inefficient. We use **Spatial SQL**.

5.1 PostGIS Architecture

PostGIS is an extension to the PostgreSQL database. It adds:

1. **Spatial Types:** Geometry (Vector) and Raster.
2. **Spatial Indexes:** R-Tree indexing for fast retrieval.
3. **Spatial Functions:** Over 300 functions (e.g., `ST_Intersection`, `ST_Buffer`).

5.2 The R-Tree Index

How does a database find "all farms within 1km" instantly? It uses an R-Tree.

- Objects are grouped into **Minimum Bounding Rectangles (MBR)**.
- Queries first check the MBR (fast integer comparison).
- Only if the MBRs intersect does the engine check the complex polygon geometry (slow floating-point math).

5.3 Practical Lab: Spatial SQL

Imagine we have two tables: `farm_plots` (Polygons) and `river_network` (Lines). We want to find the area of maize at risk of flooding (within 50m of a river).

```sql
-- Select crop type and calculate area in hectares
SELECT
    f.crop_type,
    SUM(ST_Area(
        ST_Intersection(f.geom, ST_Buffer(r.geom, 50))
    )) / 10000 AS hectares_at_risk
FROM
    farm_plots f,
    river_network r
WHERE
    f.crop_type = 'Maize'
    -- Optimization: Only check farms geometrically close to rivers
    AND ST_DWithin(f.geom, r.geom, 50)
GROUP BY
    f.crop_type;
```

Listing 2: PostGIS Query for Buffer Analysis

Code Breakdown:

- `ST_Buffer(r.geom, 50)`: Creates a 50-meter polygon "fat line" around the river.
- `ST_DWithin`: Uses the spatial index (R-Tree) to filter candidates quickly.
- `ST_Intersection`: Calculates the precise geometric overlap between the farm polygon and the river buffer.

6 Data Acquisition & Interoperability

Science requires reproducibility. We must use standard formats.

6.1 OGC Standards (Open Geospatial Consortium)

- **WMS (Web Map Service):** Returns an *image* (PNG/JPG) of the map. Good for background display. No data access.
- **WFS (Web Feature Service):** Returns the *actual vector data* (GML/GeoJSON). Allows analysis.
- **WCS (Web Coverage Service):** Returns *raster data* (pixel values). Used for downloading DEMs or Satellite bands.

Feature	Shapefile	GeoPackage
File Count	3 to 8 files (.shp, .shx, .dbf...)	1 file (.gpkg)
Max Size	2 GB	140 TB
Column Names	Max 10 characters	Unlimited
Null Support	Poor	Full

Table 1: Comparison of Geospatial File Formats

6.2 The Death of the Shapefile

The `.shp` format (ESRI) is obsolete but ubiquitous. As a PhD student, prefer **GeoPackage (.gpkg)**.

7 Advanced Topic: Georeferencing

When you capture a drone image or scan an old paper map, it has no coordinates. It exists in "Image Space" (pixels). We must transform it to "Map Space" (meters).

7.1 Affine Transformation

This is the mathematical process of scaling, rotating, and translating the image.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} A & B \\ D & E \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} C \\ F \end{bmatrix} \tag{1}$$

Where:

- (x, y): Pixel coordinates.
- (x', y'): Geographic coordinates.
- A, E: Scale factors (Pixel size).
- B, D: Rotation/Shear terms.
- C, F: Translation (Origin coordinates).

RMSE (Root Mean Square Error): The metric of georeferencing accuracy. For precision agriculture, RMSE should be < 1 pixel.

8 Review Questions

1. **Data Modeling:** You are designing a database for a national irrigation project.
   - How would you model the Canal Network? (Vector or Raster? Topology rules?)
   - How would you model the Soil Moisture Index derived from satellite? (Vector or Raster?)
2. **Geodesy:** You receive a Shapefile of farm boundaries from the Ministry of Lands (in Arc 1960) and a set of GPS points of pests from a field survey (in WGS 84).
   - If you overlay them directly, what happens?
   - What specific transformation steps must you take in QGIS or Python?
3. **Analysis:** Write pseudocode (or SQL) to find all farm plots that are larger than 2 hectares **AND** are located on a slope greater than 15 degrees. (Hint: You need to join a Vector layer with a Raster DEM).

References

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4. Iliffe, J., & Lott, R. (2008). *Datums and Map Projections: For Remote Sensing, GIS and Surveying*. Whittles Publishing.